

1

Multi-Object Disambiguation

(front/back/left/right)

 Problem	Language like “front bowl” or “bowl on the left” has no meaning to SAM3 — it returns all bowls with no ordering. Picking masks[0] silently picks the wrong one.
 When to Apply	• Scene contains ≥ 2 instances of the same object class • Task language uses front, back, middle, left, or right qualifiers
 Strategy	Detect all instances, sort by the axis implied by the qualifier (X for front/back, Y for left/right in robot frame), then select by keyword index.



Multi-Object Disambiguation

```
masks = sam3(rgb, "bowl") # returns  $\geq 2$  masks
candidates = [mask_to_world_point(m) for m in masks]

if "front" in instruction:
    candidates.sort(key=lambda p: p[0]) # sort by X
    target = candidates[0]
elif "back" in instruction:
    candidates.sort(key=lambda p: p[0], reverse=True)
    target = candidates[0]
elif "left" in instruction:
    candidates.sort(key=lambda p: p[1]) # sort by Y
    target = candidates[0]
```

2

SAM3 Prompting

(per-object prompt registry)

3

Confidence filtering

(reject low-score centroids) ...



Localization